

$$\textcircled{1} \{1, 2, 3\}$$

$$\text{Average} = \frac{\text{Sum of data}}{\text{Total no. of data}}$$

$$= \frac{1+2+3}{3}$$

$$= \frac{6}{3} = 2 //$$

$$\textcircled{2} \quad \{1+1, 2+1, 3+1\}$$

$$\Rightarrow \{2, 3, 4\}$$

MI

$$\Rightarrow \text{Avg} = \frac{2+3+4}{3} = \frac{9}{3} = 3 //$$

$$\{1+1, 2+1, 3+1\}$$

$$\text{Avg} = \frac{(1+1) + (2+1) + (3+1)}{3}$$

M2

$$= \frac{(1+2+3) + (1+1+1)}{3}$$

$$= \frac{(1+2+3)}{3} + \frac{\cancel{3}(1)}{\cancel{3}}$$

$$\text{New Avg} = \text{old Avg} + 1$$

• NOTE :

If the value of each item is

$$\frac{\text{Incr} / \text{Subt} / X}{\div}$$

by some value P , then the average

of new group will also $\frac{\text{Incr} / \text{Subt} / X}{\div} P$

$$\cdot \cdot \cdot \underline{\{1, 2, 3\}} \Rightarrow \text{Avg} = \underline{2}$$

$$\{1\underline{-1}, 2\underline{-1}, 3\underline{-1}\} \Rightarrow \text{Avg} = 2 - 1 = 1$$

$$\{1 \times 2, 2 \times 2, 3 \times 2\} \Rightarrow \text{Avg} = 2 \times 2 = 4$$

$$\left\{ \frac{1}{3}, \frac{2}{3}, \frac{3}{3} \right\} \Rightarrow \text{Avg} = \frac{2}{3} //$$

• Formula

$$\textcircled{1} \quad \text{Avg} = \frac{\text{Sum}}{\text{total}} = \frac{\Sigma}{n}$$

$$\textcircled{2} \quad \text{Avg} = \frac{\Sigma x_i f_i}{\Sigma f_i} = \frac{\Sigma x_i h_i}{n}$$

Formula

$$\text{Avg} = \frac{\text{Sum}}{\text{total}} = \frac{\Sigma}{n}$$

$\Rightarrow$

Avg ✓

n ✓

$\Sigma = ?$

$$\Sigma = \text{Avg} \times \text{total}$$

$$\Sigma = \text{Avg} \times n$$

Q] Nine numbers are written in ascending order. The middle number is the average of the nine numbers. The average of the five larger numbers is 68 and that of five smaller numbers is 44. What is the sum of all nine numbers?

① $\underline{a_1}, \underline{a_2}, \underline{a_3}, \underline{a_4}, \underline{a_5}, \underline{a_6}, \underline{a_7}, \underline{a_8}, \underline{a_9}$
(Small) [Big]

② $a_5 = \frac{\Sigma}{n} = \frac{(a_1 + a_2 + a_3 + \dots + a_9)}{9} \rightarrow \Sigma a$

$\therefore \boxed{\Sigma a = 9 a_5}$

$$\Rightarrow \frac{a_5 + a_6 + a_7 + a_8 + a_9}{5} = 68 \quad \text{--- 1}$$

$$\Rightarrow \frac{a_1 + a_2 + a_3 + a_4 + a_5}{5} = 44 \quad \text{--- 2}$$

Add 1+2 :

$$\sum a + a_5 = 5 \times 68 + 5 \times 44$$

$$9a_5 + a_5 = 560$$

$$\boxed{a_5 = 56}$$

$$\Sigma a = 925$$

$$= 9 \times 56$$

$$= 504 //$$

Shrenik Jain — Study Simplified

Q] In a family, the age of father, mother, son and grandson are A, B, C and D respectively. Given that A-B=3, B+C=78, C+D=33 and the average age of the family is 34 years then B-C is ?

$$\textcircled{1} \quad A - B = 3$$

$$B + C = 78$$

$$C + D = 33$$

$$\textcircled{2} \quad 34 = \frac{A + B + C + D}{4} \quad \Rightarrow \quad A + B + C + D = 136$$

$$\cancel{A} + B + \cancel{C} + \cancel{D} = 136$$

$$\cancel{A} - B = 3$$

$$B + \cancel{C} = 78$$

$$\cancel{C} + \cancel{D} = 33$$

$$2B - B - C = 136 - 3 - 78 - 33$$

$$B - C = 22 //$$

Q] The average age of 24 students and the principal is 15 years. When the principal's age is excluded, the average age decreases by 1 year. What is the age of principal?

$$\textcircled{1} \quad \text{Avg} = \frac{\Sigma}{n} \Rightarrow 15 = \frac{\left(\begin{array}{c} \cancel{\Sigma \text{ of age of 24 std}} + \text{Age of P} \end{array} \right)}{25}$$

336

$$\textcircled{2} \quad 14 = \frac{\Sigma \text{ of age of 24 std}}{24}$$

$$\Rightarrow \Sigma \text{ of age of 24 std} = 14 \times 24$$

$$\textcircled{15} = \frac{\left(\cancel{\Sigma \text{ of age of 24 std}}^{336} + \text{Age of P} \right)}{25}$$

$$15 \times 25 = 336 + \text{Age of P}$$

$$\text{Age of P} = 375 - 336$$

$$= 39 //$$

Q] The average weight of 3 men A,B,C is 84 kg. Another man D joins the group and the average now becomes 80 kg. If another man E, whose weight is 3 kg more than that of D replaces A then the average weight of B,C,D and E becomes 78kg. The weight of A is

$$\checkmark \textcircled{1} \quad A + B + C = 84 \times 3$$

$$\checkmark \textcircled{2} \quad A + B + C + D = 80 \times 4$$

$$\underline{\underline{2-1}}: \quad D = 80 \times 4 - 84 \times 3 = 68$$

$$E = D + 3 = 68 + 3 = 71$$

. $B, C, D, E \rightarrow \text{Avg} = 78$

$$B + C + D + \overset{71}{\cancel{E}} = 78 \times 4 = 312$$

$$\cancel{B} + \cancel{C} + \cancel{D} = 312 - 71 = 241$$

$$A + \cancel{B} + \cancel{C} + \cancel{D} = 80 \times 4$$

$$A = 80 \times 4 - 241 = 79 //$$

(Subtract)

Q] The average weight of 24 students of section A of a class is 58 kg where as the average weight of 26 students of section B of the same class is 60.5kg .Find the average weight of all the 50 students of the class.

$W = \text{Weight}$

$$\textcircled{1} \quad \text{Avg} = \frac{\Sigma \text{ of } W \text{ of } 50 \text{ std}}{50}$$

[M]

$$\textcircled{2} \quad \text{Avg (W of A)} = \frac{\Sigma \text{ of } W \text{ of } 24 \text{ std of A}}{24}$$

$$58 = \frac{\Sigma \text{ of } W \text{ of } 24 \text{ std of A}}{24}$$

$$\rightarrow \Sigma \text{ of } W \text{ of } 24 \text{ std of } A = 58 \times 24 //$$

$$\textcircled{3} \quad \text{Avg (W of } B) = \frac{\Sigma \text{ of } W \text{ of } 26 \text{ std of } B}{26}$$

$\nearrow 60.5$

$$\Sigma \text{ of } W \text{ of } 26 \text{ std of } B = 26 \times 60.5$$

$$\text{Avg} = \frac{\Sigma \text{ of W of 50 std}}{50}$$

$$= \frac{(\Sigma \text{ of W of 24 std of A}) + (\Sigma \text{ of W of 26 std of B})}{50}$$

$$\text{Avg} = \frac{(58 \times 24) + (60.5 \times 26)}{50} = 59.3 \text{ kg} //$$

M2

1) If 2 quantities are then

$$\text{Avg} = \frac{\Sigma}{\text{total}} = \frac{(\Sigma_1) + (\Sigma_2)}{h_1 + h_2}$$

$$\Rightarrow \text{Avg} = \frac{(\text{Avg}_1 \times h_1) + (\text{Avg}_2 \times h_2)}{h_1 + h_2}$$

Q] The average weight of 24 students of section A of a class is 58 kg where as the average weight of 26 students of section B of the same class is 60.5kg. Find the average weight of all the 50 students of the class.

$$\text{Avg} = \frac{(\text{Avg}_1 \times h_1) + (\text{Avg}_2 \times h_2)}{h_1 + h_2}$$

M2

$$= \frac{58 \times 24 + (60.5 \times 26)}{24 + 26}$$

$$= 59.3 //$$

Q] If 15 kg of rice costing Rs 35 per kg is mixed with 10 kg of second variety of rice costing Rs 45 per kg then the average cost per kg of the resulting mixture is

$$\underline{\text{Avg}} = \frac{(\text{Avg}_1 \times h_1) + (\text{Avg}_2 \times h_2)}{h_1 + h_2}$$

$$= \frac{35 \times 15 + (45 \times 10)}{15 + 10}$$

$$\text{Avg} = 39 \text{ ₹/kg} //$$

Q] The average weight of a class of 15 boys and 10 girls is 38.4 kg. If the average weight of the boys is 40 kg then what is the average weight of the girls?

$$\text{Avg} = \frac{(\text{Avg}_1 \times h_1) + (\text{Avg}_2 \times h_2)}{h_1 + h_2}$$

$$38.4 = \frac{40 \times 15 + \text{Avg}_2 \times 10}{15 + 10}$$

$$\Rightarrow \text{Avg}_2 = 36 //$$

Q] Sachin Tendulkar has certain average for 9 innings. In the tenth inning, he scores 100 runs thereby increasing his average by 8 runs. His new average is

① (old) $x = \frac{\Sigma \text{ of 9 innings}}{9}$

$$\underbrace{9}_{\Sigma} \quad \underbrace{100}_{100}$$

② (New) $y = x + 8$

③ $y = \frac{\Sigma \text{ of 9 innings} + 100}{10}$

$\xrightarrow{9x}$

$\Rightarrow$

$$10y = 9x + 100$$

$$10y = 9x + 100$$

$$10[x + 8] = 9x + 100$$

$$x = 20$$

$$y = 20 + x = 28 //$$